Computational Physics Project : Many Body Physics

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The complete implementation is available at
Percolation.ipynb

Contents

1 Introduction

Percolation theory provides a minimal probabilistic framework for studying the emergence of large-scale connectivity in disordered systems. Despite its simplicity, percolation exhibits a non-trivial phase transition characterized by universal critical behavior. Applications of percolation theory range from porous media flow and electrical conduction to epidemiology and network robustness.

In this work, we perform a numerical study of *site percolation* on a two-dimensional square lattice. Using Monte Carlo simulations and cluster labeling algorithms, we compute key observables as functions of the site occupation probability p and lattice size L. The objective is to identify signatures of the percolation phase transition and estimate the critical probability p_c.

2 Model Definition

We consider an $L \times L$ square lattice. Each lattice site is independently occupied with probability p and empty with probability $1 - p$. Two occupied sites are considered connected if they are nearest neighbours (up, down, left, or right).

A *cluster* is defined as a connected component of occupied sites. The system is said to *percolate* if there exists at least one cluster that spans the lattice from the top boundary to the bottom boundary.

3 Cluster Identification

Clusters are identified using a single-pass connected-component labeling algorithm inspired by the Hoshen–Kopelman method. During a sequential scan of the lattice, labels are assigned to occupied sites based on previously visited neighbours, and label equivalences are recorded. A union–find data structure is used to efficiently resolve these equivalences.

Pseudocode

```
Initialize label array to zeros
Initialize parent array for union-find
label_counter = 1

For each site (i, j) in row-major order:
    If site is empty:
        continue
    Check occupied neighbours (up, left)
    If no occupied neighbours:
        assign new label
        label_counter += 1
    Else:
        assign minimum neighbour label
        record equivalences if neighbours have different labels

Resolve equivalence classes
Relabel lattice using resolved labels
```

4 Measured Observables

For each realization, three observables are computed.

4.1 Weighted Average Cluster Size

Excluding the percolating cluster (if present), the weighted average cluster size is defined as

$$S(p) = \frac{\sum_i s_i^2}{\sum_i s_i},$$

where s_i is the size of the i-th finite cluster. This quantity is analogous to a susceptibility and is expected to peak near the critical point.

4.2 Percolation Strength

The percolation strength measures the fraction of the system occupied by the spanning cluster:

$$P_\infty(p) = \begin{cases} \frac{s_{\text{perc}}}{L^2}, & \text{if the system percolates,} \\ 0, & \text{otherwise.} \end{cases}$$

4.3 Percolation Rate

The percolation rate is defined as the probability that a randomly generated system percolates:

$$\Pi(p) = \Pr(\text{system percolates}),$$

estimated by averaging over many independent realizations.

5 Simulation Procedure

Simulations were performed for lattice sizes $L = 8, 16, 32$. For each value of the occupation probability p, N independent lattice realizations were generated. For each realization:

- A random occupancy grid was generated.
- Clusters were labeled using the connected-component algorithm.
- Percolation was detected by checking boundary-spanning clusters.
- Observables $S(p)$, $P_\infty(p)$, and $\Pi(p)$ were recorded.

The observables were then averaged over the N realizations to obtain smooth estimates as functions of p.

6 Implementation (Code Excerpts)

Below is an illustrative excerpt showing how the weighted average cluster size and percolation strength are computed for a single realization.

```
labels, counts = np.unique(labeled_grid, return_counts=True)
cluster_sizes = dict(zip(labels, counts))
cluster_sizes.pop(0, None)

if percolating_label is not None:
    P_inf = cluster_sizes[percolating_label] / (L * L)
    cluster_sizes.pop(percolating_label)
else:
    P_inf = 0.0

if cluster_sizes:
    sizes = np.array(list(cluster_sizes.values()))
    S = np.sum(sizes**2) / np.sum(sizes)
else:
    S = 0.0
```

7 Visualization

Visualization plays a crucial role in both debugging and physical interpretation. Occupancy grids, cluster labels and percolating clusters were visualized using color-coded plots.

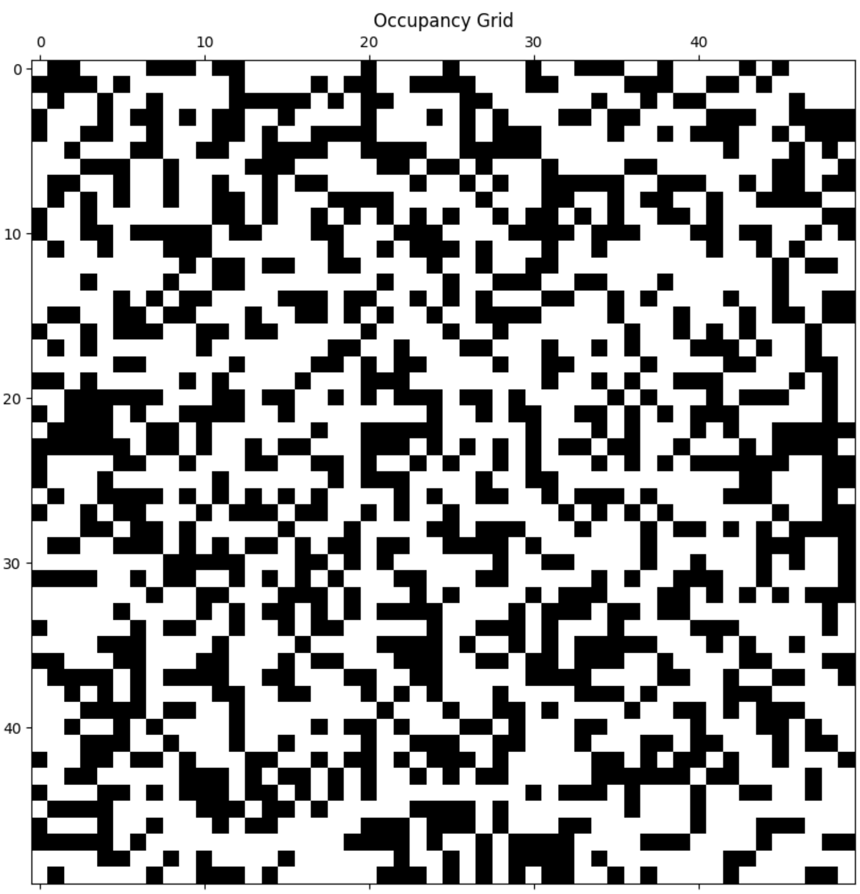


Figure 1: Example occupancy grid for a random realization.

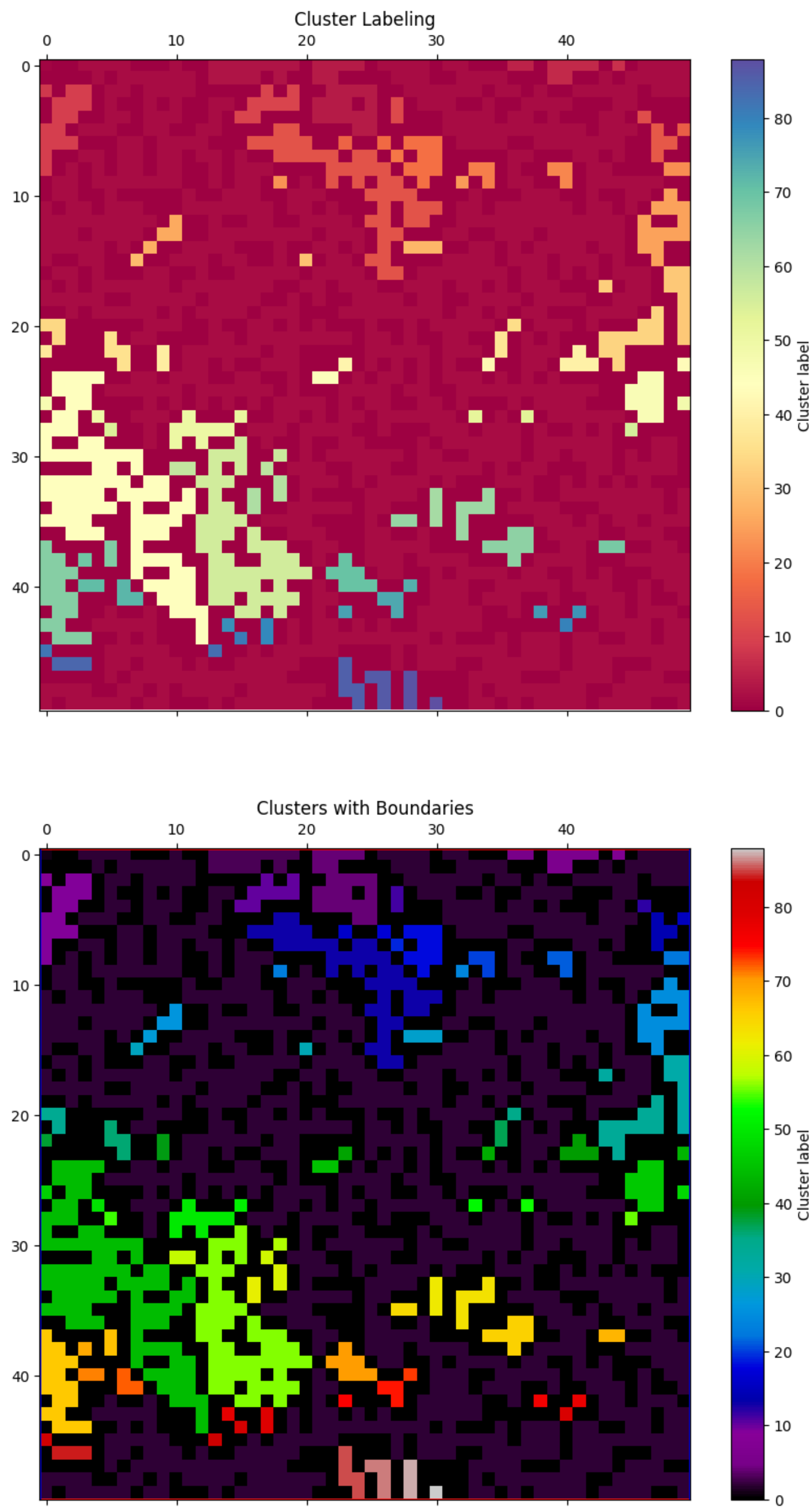


Figure 2: Cluster labeling for the same realization. Distinct colors indicate different clusters.

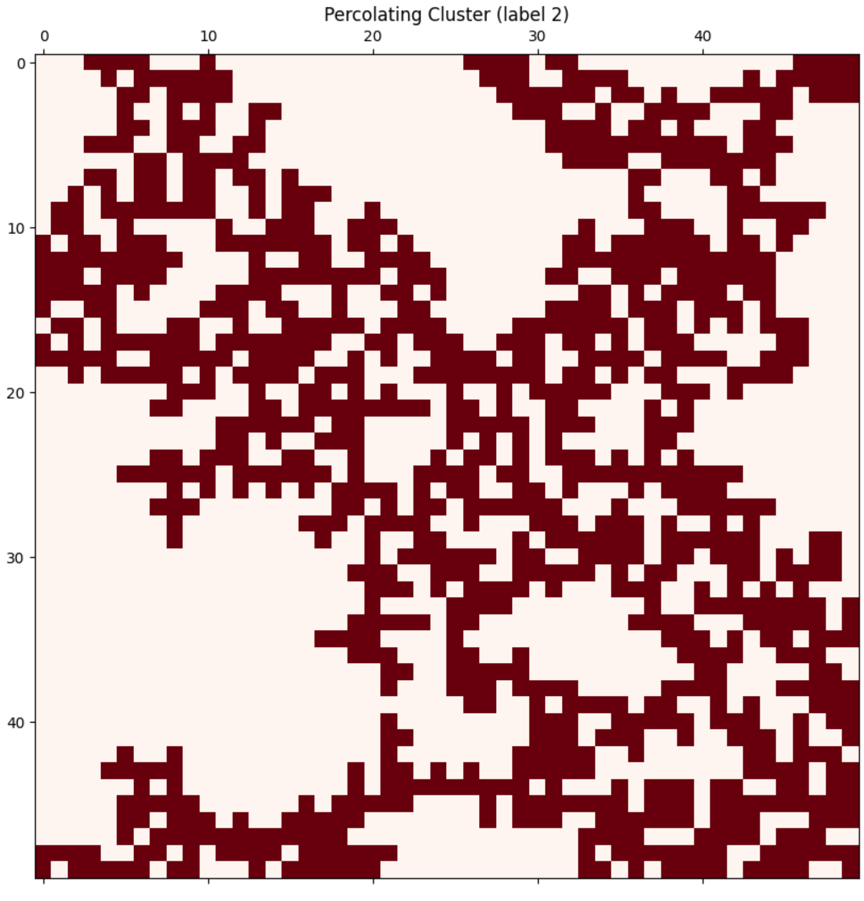
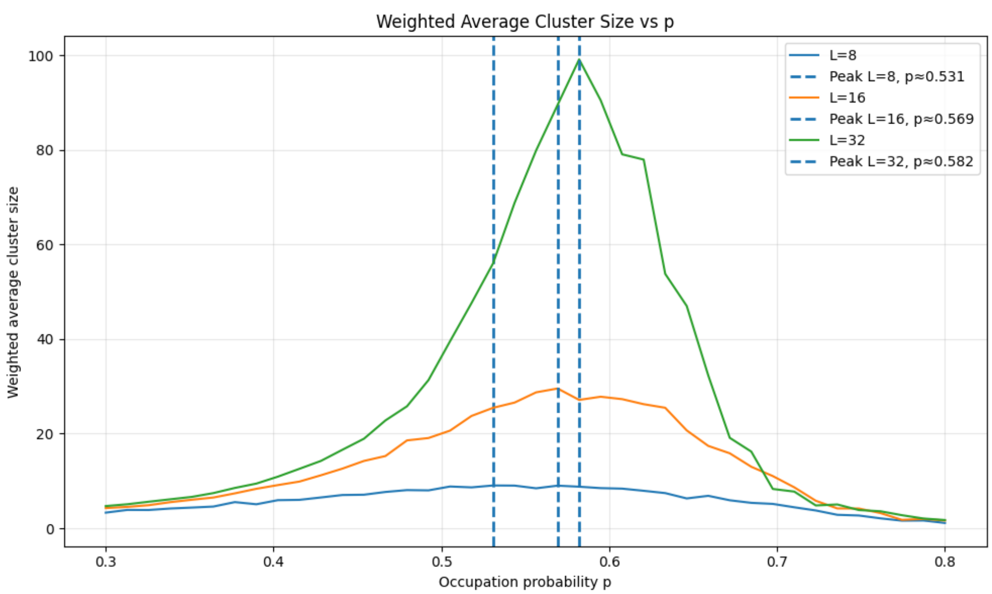


Figure 3: Percolating Cluster



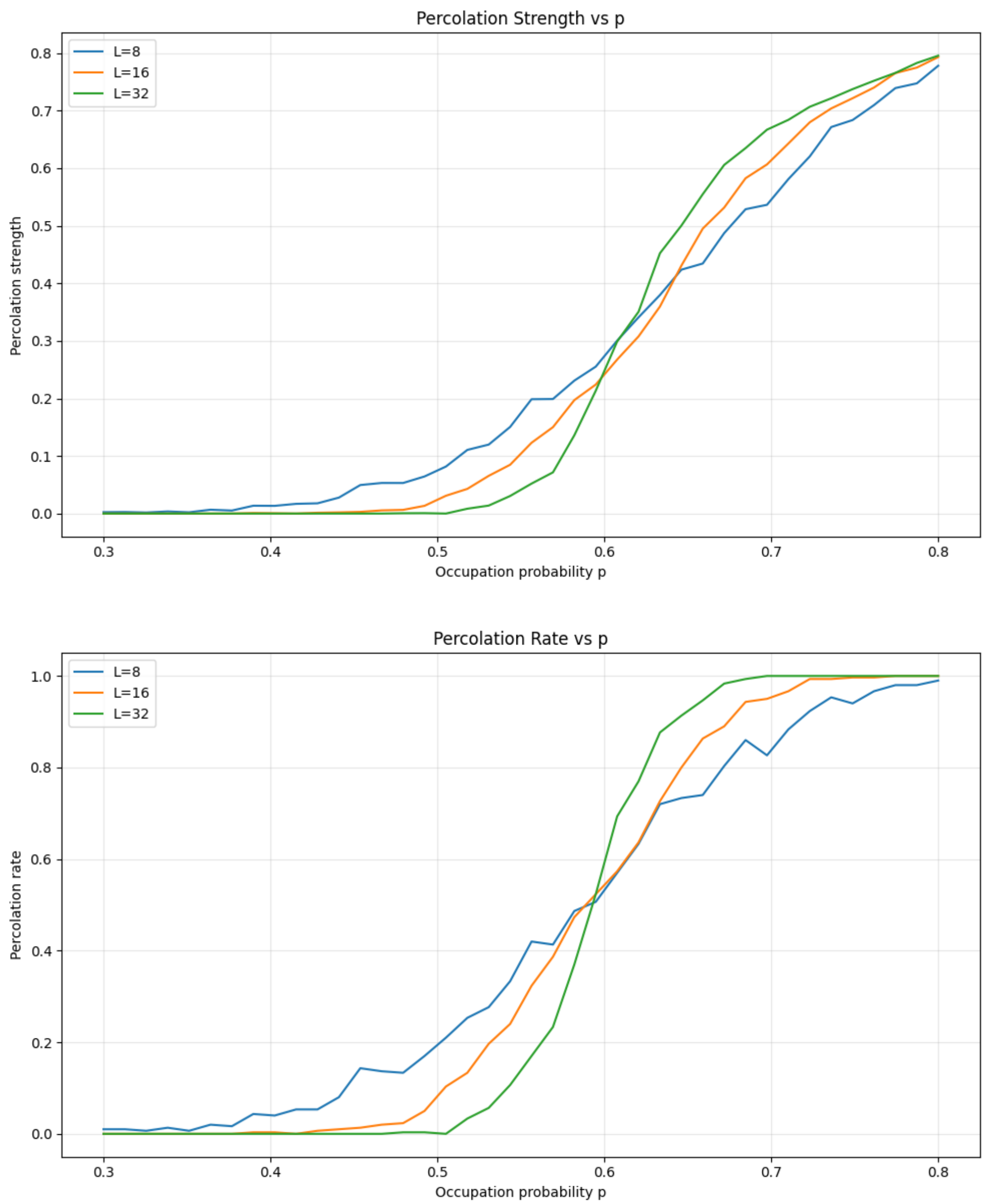


Figure 4: Weighted average cluster size, percolation strength, and percolation rate as functions of p for different lattice sizes.

8 Results and Observations

The weighted average cluster size exhibits a pronounced peak that becomes sharper and higher as the lattice size increases. This behavior is consistent with the divergence of the susceptibility in the thermodynamic limit.

The percolation strength transitions from zero to finite values near the same occupation probability. The transition becomes steeper with increasing lattice size, indicating the emergence of a macroscopic spanning cluster.

The percolation rate shows sigmoid-like behavior, with increasingly sharp transitions as L increases. The curves for different system sizes cross in a narrow region of p, providing a reliable finite-size estimator for the critical probability.

9 Conclusions

All measured observables consistently indicate a continuous percolation phase transition. As the lattice size increases, finite-size effects diminish and the estimated critical probability converges.

From the numerical results, the critical probability for site percolation on the two-dimensional square lattice is estimated to be

$$p_c \approx 0.59,$$

in excellent agreement with the known exact value $p_c \approx 0.5927$.

The results demonstrate how large-scale connectivity emerges from local randomness and highlight the role of finite-size effects in numerical studies of phase transitions.

References

- D. Stauffer and A. Aharony, *Introduction to Percolation Theory*, Taylor & Francis.
- MIT Percolation Notes: `https://www.mit.edu/~levitov/8.334/notes/percol_notes.pdf`